

METRIC MASTERY

DESCRIPTION : Teams will estimate and then measure properties of identical objects including mass, area, volume, distance, time, etc. Teams will also perform metric unit conversions.

COMPETITORS BRING/NEED :

- Writing utensil
- Simple calculator

NUMBER OF PARTICIPANTS : Maximum of 2

IMPOUND : NO

THE COMPETITION :

1. The event will be divided into 3 sections. All measurements and calculation units must be limited to the metric system.
2. Section 1 - Estimation - will involve estimating measurement properties of objects at stations. During Section 1 students will be estimating only and no measurement tools will be used.
3. The property to be estimated and the units of measure must be clearly identified in the directions at the station.
4. Part 1 will be set up as 8 individual stations. Teams will have 60 seconds per station to make their estimates.
5. Participants may not touch or feel the objects unless the station directions clearly state otherwise. (students must be allowed to lift the object if they are being required to estimate mass.)
6. Section 2 - Metric Unit Conversion - Teams will have 10 minutes to complete metric unit conversions.
7. This part must be completed after Part 1.
8. Participants will have 10 minutes to complete metric unit conversion problems to a specific specified metric unit. Conversions must be limited to the metric system only.
9. Section 3 - Measurement - will involve precisely measuring properties of objects at stations. Some of the stations may ask for calculated measurements. When this is the case, formulas will be given.
10. Measurements must be made using the supervisor-supplied instruments, expressed in the units of the instrument and as specified in the directions.
11. To receive points, the units must be given.
12. Part 3 will be a repetition of the stations from Section 1. Teams will have 120 seconds per station to make and calculate their measurements.
13. During Part 3 students will be allowed to carefully touch the objects and the measuring tools provided.

Estimations/Measurements which may be included:

- Area - 2D shapes and combined 2D shapes
 - Mass
 - Length/Distance
 - Density
 - Perimeter
 - Angles
 - Time
 - Liquid volume
 - Volume - of rectangular prisms or of other objects using water displacement
 - Surface area of 3D shapes
14. Calculated measurements are limited to area and volume of polygons and polyhedrons, density of contained liquids, solids, and powders, displacement of liquid by a solid, and mass of solids and liquids.

SCORING:

1. High Score Wins. Final score = Estimation Score + Conversion Score + Measurement Score
2. Section 1 - Estimation Score (for a total of up to 40 points)

Answers within 10% receive 5 points	Answers within 20% receive 3 points	Answers within 30% receive 1 point
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3. Section 2 - Metric Unit Conversion Score (for a total of up to 40 points)
Answers must have the metric unit included to receive points.

4. Section 3 - Measurement Score (for a total of up to 80 points)

Answers with 5% receive 10 points	Answers with 10% receive 7.5 points	Answers within 15% receive 5 points	Answers within 20% receive 2.5 points	Answers beyond 20% receive 0 points
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5. Ties will be broken by predetermined stations within Section 3 of the event.

6. Penalties, up to 10%, may be applied to teams who do not return measuring devices to their original locations, do not clean up spills, and/or intentionally alter or damage equipment or objects.